

Path 1: Pre Security > Module 6: Software Basics > Room 4: JavaScript Simple Demo

<https://tryhackme.com/room/javascriptsimpledemo>

JavaScript: Simple Demo

Explore what a basic JavaScript program looks like.

>> Task 1: Introduction

The purpose of this room is to give you an idea about one popular programming language: JavaScript. It is almost certain that you have come across the name JavaScript; however, throughout this room, we assume that you have never studied or written programs in this language.

JavaScript is used in most of the web pages that you visit on a day-to-day basis. Consequently, when we think of JavaScript engines, web browsers come to mind. In fact, JavaScript was initially developed to run on client machines, in particular, within a web browser. However, this has changed with the release of Node.js, which enables developers to build web applications with JavaScript. Consequently, as Node.js spread, JavaScript was no longer just a client-side programming language but also a server-side one.

>> Task 2: Variables

Q1) What word is used to declare a variable?

Answer: let

Q2) What word is used to declare a constant?

Answer: const

Q3) What is the method that we call to display text on the screen?

Answer: console.log()

>> Task 3: Prompting the User For Input

```
import * as readline from "node:readline/promises";
import { stdin as input, stdout as output } from "node:process";

const rl = readline.createInterface({ input, output });

try {
  const secret = Math.floor(Math.random() * (20)) + 1; // 1 <= secret <= 20
  let tries = 0;
  let guess = 0; // start with a value that cannot be the secret (since secret is 1..20)

  console.log("I'm thinking of a number between 1 and 20");

  const text = await rl.question("Take a guess: "); // rl.question() returns text (a
string)
  guess = parseInt(text, 10); // convert the text to a number

  tries = tries + 1; // add 1 try

} finally {
  rl.close();
}
```

Q1) What method is used to convert user input into a number?

Answer: parseInt()

>> Task 4: Conditional Statements

```
import * as readline from "node:readline/promises";
import { stdin as input, stdout as output } from "node:process";

const rl = readline.createInterface({ input, output });

try {
  const secret =
    Math.floor(Math.random() * (20)) + 1; // 1 <= secret <= 20
  let tries = 0;
  let guess = 0; // start with a value that cannot be the secret (since secret is 1..20)

  console.log("I'm thinking of a number between 1 and 20");

  const text = await rl.question("Take a guess: "); // rl.question() returns text (a
string)
  guess = parseInt(text, 10); // convert the text to a number

  tries = tries + 1; // add 1 try

  // Give a hint using if / else if / else.
  if (guess < 1 || guess > 20) {
    console.log("That number is out of range. Try again.");
  } else if (guess < secret) {
    console.log("Too low, try again.");
  } else if (guess > secret) {
    console.log("Too high, try again.");
  } else {
    console.log("You got it in", tries, "tries!");
  }
} finally {
  rl.close();
}
```

Q1) The secret is 10. What will our program display on the screen if the user makes a guess of 15?

Answer: Too high, try again.

Q2) The secret is 10. What will our program display on the screen if the user makes a guess of 35?

Answer: That number is out of range. Try again.

>> Task 5: Iterations

```
import * as readline from "node:readline/promises";
import { stdin as input, stdout as output } from "node:process";

const rl = readline.createInterface({ input, output });

try {
  const secret =
    Math.floor(Math.random() * (20)) + 1; // 1 <= secret <= 20
  let tries = 0;
  let guess = 0; // start with a value that cannot be the secret (since secret is 1..20)

  console.log("I'm thinking of a number between 1 and 20");

  // Repeat until the user guesses the secret number.
  while (guess !== secret) {
    const text = await rl.question("Take a guess: "); // rl.question() returns text (a
string)
    guess = parseInt(text, 10); // convert the text to a number

    tries = tries + 1; // add 1 try

    // Give a hint using if / else if / else.
    if (guess < 1 || guess > 20) {
      console.log("That number is out of range. Try again.");
    } else if (guess < secret) {
      console.log("Too low, try again.");
    } else if (guess > secret) {
      console.log("Too high, try again.");
    } else {
      console.log("You got it in", tries, "tries!");
    }
  }
} finally {
  rl.close();
}
```

Q1) What is the name of the loop that we used in this task?

Answer: while

Q2) What is the name of the variable that is incremented by one when the user makes a new wrong guess?

Answer: tries

Q3) How is “not equal” written in JavaScript?

Answer: !==